Experiment 14

Aliasing and Anti-Aliasing Filter Analysis

OBJECTIVES

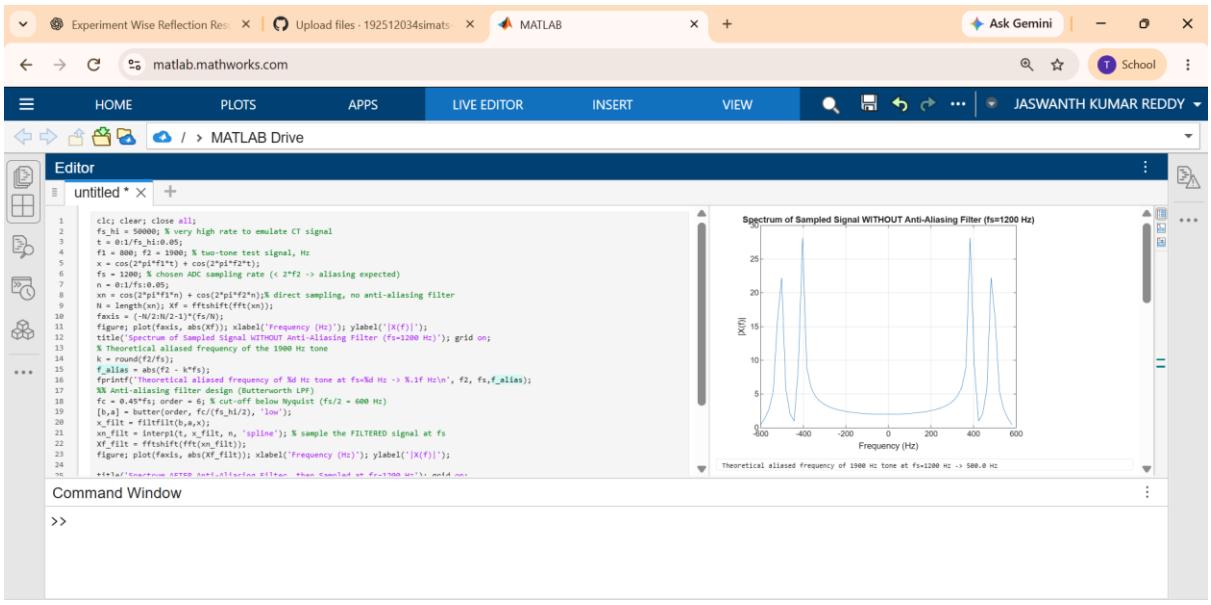
• Demonstrate aliasing when a signal is sampled below its Nyquist rate and compute the aliased frequency.

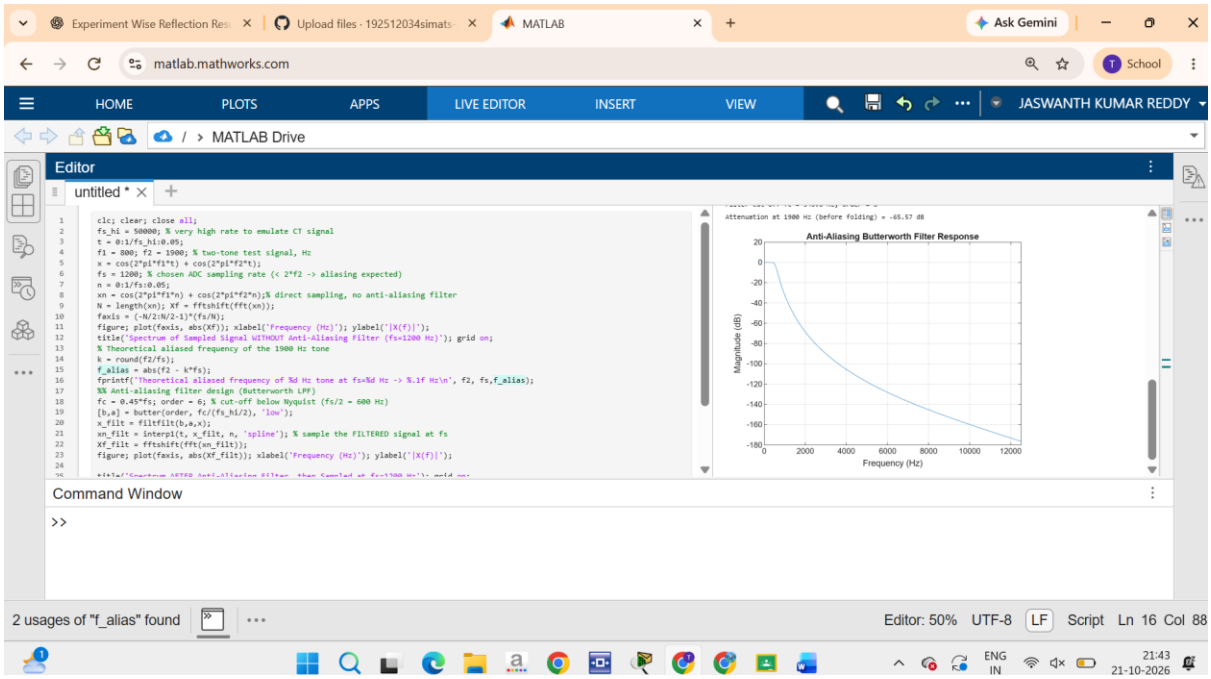
• Design an anti-aliasing low-pass Butterworth filter with an appropriate cut-off frequency and attenuation.

• Verify that filtering before sampling reduces the out-of-band energy responsible for aliasing.

• Relate anti-aliasing to real-time acquisition front ends such as ADAS sensor and camera pipelines.

OUTPUT:





Result: Aliasing effects were analyzed by varying the sampling frequency, and the role of anti-aliasing filters in reducing unwanted frequency components was studied.